Emily Zhang

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EDUCATION

McGill University **Expected 2027**
BSc. in Computer Science and Artificial Intelligence
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Design, Computer Architecture, Artificial Intelligence, Applied Machine Learning, Natural Language Processing, Databases, Statistics, Probability

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, Go, Rust, SQL, HTML/CSS
Frameworks & Tools: LangChain, PySpark, Qt, Node.js, Next.js, Git, Docker, Kubernetes, Databricks, Azure DevOps, CI/CD
Databases & Cloud: Qdrant, PostgreSQL, MySQL, Firebase/Firestore, AWS

WORK EXPERIENCE

Google ***Waterloo, ON***
Software Developer Intern *May 2025 – Aug 2025*
- Reduced color calibration preprocessing errors by 50% across varying lighting conditions for Gemini-powered AI glasses by developing a Python statistical algorithm for automated pattern detection in noisy sensor data.
- Doubled API verification throughput for Android XR smart glasses by building an automated display API validation pipeline integrating ADB device orchestration, hardware data processing, and quantitative testing.
- Reduced hardware and calibration debugging time by 50% by implementing a Protobuf-based CLI diagnostic tool that automated device-data processing, report generation, and visualization.

Autodesk ***Montreal, QC***
Software Developer Intern *Sept 2024 – Dec 2024*
- Implemented OpenUSD light-linking support in Maya and 3ds Max using C++, Python, and Qt, enabling native light-to-object relationship management in USD scenes and reducing manual setup effort by 40%.
- Developed a Qt-based light selection widget with type-ahead search and USD path handling, reducing light-assignment time by 60% in large production scenes.

CIBC ***Toronto, ON***
AI/ML Software Developer Intern
- Developed a production RAG platform with an MCP integration layer using LangChain and Qdrant, combining vector search, BM25, and reranking to improve retrieval relevance by 20% across internal knowledge sources.
- Implemented a Python-based mortgage validation engine on Databricks PySpark, and Delta Lake, orchestrating AI-based document classification to automate complex compliance checks and reduce manual review effort by 30%.

McGill University ***Montreal, QC***
Undergraduate Research Assistant *(https://atrehab.ca/)*
- Developed a Python biomechanical data pipeline for motion-capture and IMU sensor data, automating data ingestion, signal processing, 3D movement reconstruction, and upper-body classification to support post-stroke motor rehabilitation.

PROJECTS

Personal website: https://emilyzzzhang.github.io/

Paws in Peril | *Python, Design Patterns, OOP, pytest* | **Github**
- Developed a multiplayer 2D pixel animal-rescue game with Dijkstra-based NPC pathfinding and core gameplay systems; awarded Best Course Project in Software Design.

HexaHaven | *TypeScript, Phaser 3, Vite, Node.js, Express, Socket.IO, Tailwind CSS, Firebase/Firestore* | **Github**
- Engineered a multiplayer hex-strategy game of resource collection, trades, and settlement construction, with procedural terrain from multi-octave fBm gradient noise and argmax biome selection over weighted linear combinations.

InnerSpace | TypeScript, Next.js, Prisma/PostgreSQL, Zod, Google Gemini API | **Github**
- Developed an AI-powered journaling web app with voice and text input, Gemini-generated follow-up prompts, and automated extraction of moods, summaries, and actionable insights.

EXTRACURRICULARS

- **Hackathon Director, McWICs |** Organized and led technical workshops for Hack McWICS, engaging ~200 participants.
- **Tutor, Girls Who Code** | Taught Python and core programming concepts to a class of high school students weekly.